

Memebox by Solidly Labs Audit Summary

Prepared by Cyfrin

Version 2.2

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Solidly Labs on Memebox by Solidly Labs. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.2 dated May 4, 2024, was delivered to Solidly Labs, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 6 days, the Cyfrin team conducted an audit on the Memebox by Solidly Labs code provided by Solidly Labs. The findings consist of 1 Medium & 3 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Memebox by Solidly Labs
Repository	v2-core
Commit	757b18ad0578...
Audit Timeline	Apr 22nd - Apr 29th, 2024
Methods	Manual Review, Stateful Fuzzing

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	1	1	0
Low Risk	3	0	3
Informational	5	0	5
Gas Optimizations	1	0	1
Total	10	1	9

6 Audit Scope

Cyfrin conducted an audit of Solidly V2 Memecore based on the code present in the repository commit hash 757b18a, with remediations applied to the final commit hash 14533e7.

All contracts within the flattened SolidlyV2-memecore.sol file were included in the scope of the audit, including:

- Math
- SolidlyV2Accounting
- SolidlyV2LockBox
- SolidlyV2Pair
- SolidlyV2ERC42069
- SolidlyV2Factory

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Solidly V2 Memecore is a new smart contract developed by Solidly Labs and in use on the Memebox platform. It is a renewed interpretation and reimplementation of the xy >= k AMM written in Solidity version 0.8.19, specifically tailored to cater to memecoins.

Key features include:

- LP fee abstraction and configurable fee structure.
- Vertically-integrated locking mechanism & native memecoin functionality (e.g. locks, burns, split, extends, transfers).
- The ability to earn fees in perpetuity even after liquidity is burned.
- Lockers/burners can access their fees at any time.
- Laying the groundwork for yield tokenization by making the yield receipt fungible.

9 Attestation

Cyfrin attests that it performed a security review of the Memebox by Solidly Labs code provided by Solidly Labs.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2024-05-04-cyfrin-solidly-v2-memecore-v2-2.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Solidly Labs has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=b88ff6aec943cdeb2b9df44749b31315c88bd74347234869cc78b840856bfc39
F=2024-05-04-cyfrin-solidly-v2-memecore-v2-2-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.